

Lab Report 1

Course Code : CSE0611324
Course Title : Microprocessors, Microcontrollers and Assembly Language Lab

Task : Print the sample output

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Dept. : CSE
Batch : 55
Section : 6B1

Submission Date : 09.08.2026

Experiment No: 1

Experiment Name:

Program should print the given sample output

Introduction:

In this lab we learnt the basic syntax of Assembly language. We stored some values in AX register temporarily, then moved them to permanent BX and finally displayed output using DX registers. All that was wrapped up with a practice problem that included us printing output as the sample provided.

Given Assignment / Sample Output:



Lab Report 1

Abrar Mahbub Tanim • 11:31 AM

100 points | Due Aug 9

Your program should print the given sample output:

Input:

1

9

Output:

1

9

Code:

.model small	; PRINT NEW LINE	; PRINT NEW LINE
.stack 100h		
	mov ah, 2	mov ah, 2
.data ; DATA SEGMENT	mov dl, 10	mov dl, 10
var1 db 'Input:\$'	int 21h	int 21h
var2 db 'Output:\$'		
	; CARRY BACK	; CARRY BACK
.code		
main proc	mov ah, 2	mov ah, 2
	mov dl, 13	mov dl, 13
	int 21h	int 21h
mov ax, @data		
mov ds,ax	; INPUT 9	; OUTPUT 1
; PRINT "INPUT"	mov ah, 1	mov ah,2
	int 21h	mov dl, bl
mov dx, offset var1	mov bh, al	int 21h
mov ah, 09h		
int 21h	; PRINT NEW LINE	; PRINT NEW LINE
; PRINT NEW LINE	mov ah, 2	mov ah, 2
	mov dl, 10	mov dl, 10
mov ah, 2	int 21h	int 21h
mov dl, 10		
int 21h		; CARRY BACK
	; PRINT NEW LINE	
; CARRY BACK		
	mov ah, 2	mov ah, 2
mov ah, 2	mov dl, 10	mov dl, 13
mov dl, 13	int 21h	int 21h
int 21h		; OUTPUT 9
	; CARRY BACK	
; INPUT 1		
	mov ah, 2	mov ah,2
mov ah, 1	mov dl, 13	mov dl, bh
int 21h	int 21h	int 21h
mov bl, al		
	; PRINT "OUTPUT"	exit:
		mov ah,4ch
mov dx, offset var2		int 21h
mov ah, 09h		main endp
int 21h		end main

Output:

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```
010 main proc
011
012     mov ax, @data
013     mov ds, ax
014
015     ; PRINT "INPUT"
016
017     mov dx, offset var1
018     mov ah, 09h
019     int 21h
020
021     ; PRINT NEW LINE
022
023     mov ah, 2
024     mov dl, 10
025     int 21h
026
027     ; CARRY BACK
028
029     mov ah, 2
030     mov dl, 13
031     int 21h
032
033     ; INPUT 1
034
035     mov ah, 1
036     int 21h
037     mov bl, al
038
039     ; PRINT NEW LINE
040
041     mov ah, 2
042     mov dl, 10
043     int 21h
044
045     ; CARRY BACK
046
047     mov ah, 2
048     mov dl, 13
049     int 21h
050
051     ; INPUT 9
052
053     mov ah, 1
054     int 21h
055     mov bh, al
056
057     ; PRINT NEW LINE
058
059     mov ah, 2
060     mov dl, 10
061     int 21h
```

emulator screen (80x25 chars)

Input:
1
9

Output:
1
9

emulator: A_1.exe_

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Load reload step back single step run step delay ms: 0

registers		F400:0200		F400:0204	
	H L				
AX	4C 39	F4200:	FF 255 RES	BIOS DI	
BX	39 31	F4201:	FF 255 RES	INT 021h	
CX	01 81	F4202:	CD 205 =	IRET	
DX	00 39	F4203:	21 033 1	ADD CBX + SI, AL	
CS	F400	F4204:	CF 207 1	ADD CBX + SI, AL	
IP	0204	F4205:	00 000 NULL	ADD CBX + SI, AL	
SS	0710	F4206:	00 000 NULL	ADD CBX + SI, AL	
SP	00FA	F4207:	00 000 NULL	ADD CBX + SI, AL	
BP	0000	F4208:	00 000 NULL	ADD CBX + SI, AL	
SI	0000	F4209:	00 000 NULL	ADD CBX + SI, AL	
DI	0000	F420A:	00 000 NULL	ADD CBX + SI, AL	
DS	0720	F420B:	00 000 NULL	ADD CBX + SI, AL	
ES	0700	F420C:	00 000 NULL	ADD CBX + SI, AL	
		F420D:	00 000 NULL	ADD CBX + SI, AL	
		F420E:	00 000 NULL	ADD CBX + SI, AL	
		F420F:	00 000 NULL	ADD CBX + SI, AL	
		F4210:	00 000 NULL	ADD CBX + SI, AL	
		F4211:	00 000 NULL	ADD CBX + SI, AL	
		F4212:	00 000 NULL	ADD CBX + SI, AL	
		F4213:	00 000 NULL	ADD CBX + SI, AL	
		F4214:	00 000 NULL	...	

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```
113 ; OUTPUT 9
114
115     mov ah, 2
116     mov dl, bh
117     int 21h
118
119
120
121
122 exit:
123     mov ah, 4ch
124     int 21h
```

message

PROGRAM HAS RETURNED CONTROL TO THE OPERATING SYSTEM

OK

line: 23 col: 58 drag a file here to open

Conclusion:

We successfully implemented the sample output.